

Weak separation and finite-trace coding: independence of infinite fin-intersecting MAD families

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Abstract

We prove that, if $\mathfrak{s} < \mathfrak{ap}$, an almost disjoint family is fin-intersecting exactly when its cardinality is less than $\mathfrak{s}$. Consequently, no infinite fin-intersecting maximal almost disjoint family exists in this region. The main lemma converts countably many weak intersection tests on a family of size less than $\mathfrak{b}$ into eventual traces of one sequence of disjoint nonempty finite sets. Weak separation then encodes a splitting family and its complements on prescribed members of any sufficiently large almost disjoint family. Combined with a consistency theorem of Banach, Machura and Zdomskyy and the positive results of Corral and Rodrigues, this yields the relative independence over ZFC of the existence of an infinite fin-intersecting MAD family. The negative model also contains a MAD family with pseudocompact Vietoris hyperspace. A companion Lean development checks the exact obstruction, the comparison $\mathfrak{ap} \leq \mathfrak{b}$, and the CH existence construction; the forcing models and the full ZFC independence theorem are not formalized there.

1 Introduction

An almost disjoint family can have centered traces along an infinite subsequence of every sequence of disjoint finite blocks. Corral and Rodrigues call this property *fin-intersecting* and use it to obtain MAD families whose associated Vietoris hyperspaces are pseudocompact [1]. Their Question 3.7 asks whether ZFC proves the existence of a fin-intersecting MAD family. Throughout this paper, a MAD family is required to have infinitely many members.

Let $\mathbf{E}$ denote the assertion that an infinite fin-intersecting MAD family on ω exists. The main combinatorial result is the following.

Theorem 1.1. *Assume $\mathfrak{s} < \mathfrak{ap}$. For every almost disjoint family $\mathcal{A} \subseteq [\omega]^\omega$,*

$$\text{FI}(\mathcal{A}) \iff |\mathcal{A}| < \mathfrak{s}.$$

In particular, $\neg \mathbf{E}$ holds.

The positive direction of the equivalence is the known small-family bound of Corral and Rodrigues. Our argument for the converse starts with an arbitrary candidate family. We choose pairs of distinct members indexed by a splitting family and prescribe complementary traces on each pair. Weak separation supplies countably many infinite testing sets. A single eventual

bound converts these tests into disjoint finite blocks. Any infinite set of indices is split by one of the labels, giving two infinite retained traces whose intersection is finite.

The inequality $\mathfrak{ap} \leq \mathfrak{b}$ is the external cardinal comparison that makes this coding possible. Banach, Machura and Zdomskyy give a relative consistency model with $\mathfrak{s}_\omega = \aleph_1 < \mathfrak{dp} = \mathfrak{c} = \aleph_2$ [2]. It satisfies $\mathfrak{s} < \mathfrak{ap}$, so Theorem 1.1 rules out every infinite FI MAD family there. On the other hand, Corral and Rodrigues prove $\mathfrak{ap} = \mathfrak{s} = \mathfrak{c} \Rightarrow \mathsf{E}$, in particular $\mathsf{CH} \Rightarrow \mathsf{E}$. Thus, with consistency understood syntactically,

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZFC} + \mathsf{E}) \wedge \text{Con}(\text{ZFC} + \neg\mathsf{E}).$$

Sections 3–4 contain the new coding and exclusion arguments. Section 5 gives a self-contained reconstruction of the known CH construction. Section 6 states the model inputs precisely and derives independence. We also separate FI existence from the existence of a MAD family with pseudocompact hyperspace. Section 8 records exactly what the companion Lean files prove.

2 Definitions and preliminary facts

We work in ZFC. Write $A \subseteq^* B$ when $A \setminus B$ is finite and $A =^* B$ when their symmetric difference is finite. For functions on ω , $f \leq^* g$ means that $f(n) \leq g(n)$ for all sufficiently large n . We use $[0, k) = \{0, \dots, k-1\}$.

Definition 2.1. A family $\mathcal{A} \subseteq [\omega]^\omega$ is *almost disjoint* (AD) if distinct members have finite intersection. It is *MAD* if $\mathcal{A}$ is infinite and

$$\forall X \in [\omega]^\omega \exists A \in \mathcal{A} \quad |X \cap A| = \aleph_0.$$

Finite or empty AD families are allowed in intermediate arguments, but do not count as MAD families. A *fin-sequence* is a sequence $C = (C_n : n < \omega)$ of pairwise disjoint nonempty finite subsets of ω . Its trace on A is

$$\text{tr}_C(A) = \{n : A \cap C_n \neq \emptyset\}.$$

There is no uniform bound on the block sizes, and the blocks need not cover ω .

Definition 2.2 ([1, Definition 2.3]). A family of infinite sets is *centered* if every nonempty finite subfamily has infinite intersection. The empty family is centered. An AD family $\mathcal{A}$ is *fin-intersecting* (FI) if, for every fin-sequence C , some infinite $I \subseteq \omega$ makes

$$\{I \cap \text{tr}_C(A) : A \in \mathcal{A}, |I \cap \text{tr}_C(A)| = \aleph_0\}$$

centered. The test applies to *all* infinite retained traces, after finite traces have been discarded.

The splitting number $\mathfrak{s}$ is the least size of a family $\mathcal{S}$ of subsets of ω such that every infinite I is split by some $S \in \mathcal{S}$: both $I \cap S$ and $I \setminus S$ are infinite. The bounding number $\mathfrak{b}$ is the least size of an unbounded family in $(\omega^\omega, \leq^*)$. Let $\mathfrak{a}$ be the least size of an infinite MAD family, and let $\mathfrak{c} = 2^{\aleph_0}$.

For $\mathcal{B} \subseteq \mathcal{A}$, a set $X \subseteq \omega$ *weakly separates* $\mathcal{B}$ from $\mathcal{A} \setminus \mathcal{B}$ if

$$|A \cap X| = \aleph_0 \quad (A \in \mathcal{B}), \quad |A \cap X| < \aleph_0 \quad (A \in \mathcal{A} \setminus \mathcal{B}).$$

The invariant $\mathfrak{ap}$ is the least size of an AD family with a subfamily that cannot be weakly separated from its complement. This definition uses infinite versus finite intersection, not almost containment.

We use the standard inequalities

$$\aleph_1 \leq \mathfrak{ap} \leq \mathfrak{b} \leq \mathfrak{a} \leq \mathfrak{c}, \quad \aleph_1 \leq \mathfrak{s} \leq \mathfrak{c}. \quad (1)$$

In particular, $\mathfrak{ap} \leq \mathfrak{b}$ follows from [2, Theorem 2]. For completeness we recall the diagonal argument for $\mathfrak{b} \leq \mathfrak{a}$.

Lemma 2.3. *An infinite AD family of size less than $\mathfrak{b}$ is not MAD.*

Proof. Choose distinct $A_i \in \mathcal{A}$ and put $E_i = A_i \setminus \bigcup_{j < i} A_j$. The sets E_i are infinite and pairwise disjoint. For each $A \in \mathcal{A}$, let

$$f_A(i) = \begin{cases} 1 + \max(\{0\} \cup (A \cap E_i)), & A \neq A_i, \\ 0, & A = A_i. \end{cases}$$

Choose a common eventual bound g and choose $x_i \in E_i$ with $x_i \geq g(i)$. The points are distinct. For a fixed A , all sufficiently late x_i lie outside A , except possibly the unique index with $A = A_i$. Thus $\{x_i : i < \omega\}$ is infinite and almost disjoint from every member. $\square$

Lemma 2.4. *The FI property passes to subfamilies. Every AD family of size less than $\mathfrak{s}$ is FI.*

Proof. For heredity, every retained trace for a subfamily is a retained trace for the original family. For the second assertion, fix C . Its trace family has size less than $\mathfrak{s}$, so it fails to split some infinite I . Each trace is finite or cofinite in I . Every retained trace is therefore cofinite in I , and the retained family is centered. This is the argument of [1, Theorem 3.1]. $\square$

3 Uniform finite-trace coding

The following lemma makes no almost-disjointness assumption. Its point is to use one eventual bound before choosing any of the block endpoints.

Lemma 3.1 (Uniform coding). *Let $\mathcal{U} \subseteq \mathcal{P}(\omega)$ have size less than $\mathfrak{b}$, and assign $S_A \subseteq \omega$ to each $A \in \mathcal{U}$. Suppose that infinite sets $X_n \subseteq \omega$ satisfy*

$$|A \cap X_n| = \aleph_0 \iff n \in S_A.$$

There is a fin-sequence C such that $\text{tr}_C(A) = {}^ S_A$ for every $A \in \mathcal{U}$.*

Proof. First bound each negative test. Set

$$d_A(k) = \begin{cases} 0, & k \in S_A, \\ 1 + \max(\{0\} \cup (A \cap X_k)), & k \notin S_A. \end{cases}$$

Thus, in the negative case, $A \cap X_k \subseteq [0, d_A(k))$. For a positive test and an endpoint k , the infinite set $A \cap X_i$ has a least element at least k . Collect the finitely many relevant witnesses:

$$P_A(k) = \{\min(A \cap X_i \cap [k, \infty)) : i \leq k, i \in S_A\}, \quad h_A(k) = \max(\{k+1, d_A(k)\} \cup P_A(k)).$$

Choose $g \in \omega^\omega$ with $h_A \leq^* g$ for every $A \in \mathcal{U}$, increasing it if necessary so that $g(k) \geq k + 1$. This is the sole use of $|\mathcal{U}| < \mathfrak{b}$.

Define the endpoints recursively. Let $r_0 = 0$ and, given r_n , put

$$\begin{aligned} \ell_n &= \max\{n, g(n), r_n\}, \\ z_n &= \min(X_n \cap [\ell_n, \infty)), \\ r_{n+1} &= 1 + \max\{\ell_n, g(\ell_n), z_n\}, \quad C_n = X_n \cap [\ell_n, r_{n+1}). \end{aligned}$$

Every z_n exists and belongs to C_n , so the blocks are nonempty and finite. Since $\ell_{n+1} \geq r_{n+1}$, the successive windows, and hence all the blocks, are disjoint.

Fix A and choose N_A so that $h_A(k) \leq g(k)$ for every $k \geq N_A$. For $n \geq N_A$ with $n \notin S_A$, we have

$$A \cap X_n \subseteq [0, d_A(n)) \subseteq [0, g(n)) \subseteq [0, \ell_n),$$

so $A \cap C_n = \emptyset$. If $n \in S_A$, then $n \leq \ell_n$ and $\ell_n \geq N_A$. The witness for test n occurs in $P_A(\ell_n)$, giving

$$\ell_n \leq \min(A \cap X_n \cap [\ell_n, \infty)) \leq h_A(\ell_n) \leq g(\ell_n) < r_{n+1}.$$

It lies in $A \cap C_n$. Thus membership in $\text{tr}_C(A)$ and S_A agrees from N_A onward, as required. $\square$

Remark 3.2. All functions h_A are defined before g and the endpoints are chosen. The later evaluation at ℓ_n is legitimate because $\ell_n \geq n$. The thresholds N_A can depend on A ; no uniform threshold over the whole family is asserted. Any choices of positive witnesses in place of the minima also work, which is the formulation used in Lean.

Corollary 3.3 (Arbitrary labels on a small infinite AD family). *If $\mathcal{A}$ is infinite, AD, and $|\mathcal{A}| < \mathfrak{ap}$, then every assignment $A \mapsto S_A \subseteq \omega$ is realized modulo finite by the traces of one fin-sequence.*

Proof. For each n , put $\mathcal{B}_n = \{A \in \mathcal{A} : n \in S_A\}$. If $\mathcal{B}_n$ is nonempty, weak separation provides a testing set X_n ; its infinite intersection with any positive member makes X_n infinite. If $\mathcal{B}_n$ is empty, Lemma 2.3 and $|\mathcal{A}| < \mathfrak{ap} \leq \mathfrak{b}$ provide an infinite set almost disjoint from all of $\mathcal{A}$; use it as X_n . Apply Lemma 3.1. $\square$

Infinitude of the family in this corollary matters: a finite partition of ω cannot have all its traces eventually empty. In the main argument below every positive fiber is nonempty, so this special case can also be avoided.

4 Excluding every sufficiently large candidate

Proof of Theorem 1.1. The forward obstruction is the only new direction. Let $\mathcal{M}$ be any AD family with $|\mathcal{M}| \geq \mathfrak{s}$. Since $2\mathfrak{s} = \mathfrak{s}$, choose distinct members

$$\mathcal{A} = \{a_\xi^0, a_\xi^1 : \xi < \mathfrak{s}\} \subseteq \mathcal{M}.$$

Take a splitting family $(S_\xi : \xi < \mathfrak{s})$ and label the selected members by $S_{a_\xi^0} = S_\xi$ and $S_{a_\xi^1} = \omega \setminus S_\xi$. Corollary 3.3 gives a fixed fin-sequence C with these traces modulo finite. Equivalently, one can weakly separate, for each n , the nonempty subfamily

$$\{a_\xi^0 : n \in S_\xi\} \cup \{a_\xi^1 : n \notin S_\xi\}$$

and apply Lemma 3.1 directly using $\mathfrak{s} < \mathfrak{ap} \leq \mathfrak{b}$.

Let I be any infinite set of indices, and choose S_ξ splitting it. The finite errors

$$D_0 = \text{tr}_C(a_\xi^0) \triangle S_\xi, \quad D_1 = \text{tr}_C(a_\xi^1) \triangle (\omega \setminus S_\xi)$$

do not destroy infinitude of either restricted trace. Both therefore survive the deletion of finite traces. But

$$\text{tr}_C(a_\xi^0) \cap \text{tr}_C(a_\xi^1) \subseteq D_0 \cup D_1,$$

so their intersection is finite. They contradict centeredness, and I cannot witness FI for C . The same C works for every infinite I . Thus $\mathcal{M}$ is not FI. Lemma 2.4 supplies the opposite implication.

Finally, any infinite MAD family has size at least $\mathfrak{a} \geq \mathfrak{b} \geq \mathfrak{ap} > \mathfrak{s}$, so it is excluded by the same argument. $\square$

Corollary 4.1. *ZFC proves $\mathsf{E} \Rightarrow \mathfrak{ap} \leq \mathfrak{s}$.*

The quantifiers in the negative conclusion are $\forall \mathcal{M} \exists C \forall I$: the blocking sequence can depend on the candidate family. This neither changes its members nor merely constructs one special non-FI MAD family.

5 The known positive direction under CH

Corral and Rodrigues prove the stronger positive theorem under $\mathfrak{ap} = \mathfrak{s} = \mathfrak{c}$ [1, Theorem 3.4]. We reconstruct the CH case to make the two sides of the independence argument explicit; no novelty is claimed for this positive direction.

For an AD family $\mathcal{A}$, let

$$\mathcal{I}(\mathcal{A}) = \{Y \subseteq \omega : \exists F \in [\mathcal{A}]^{<\omega}, Y \subseteq^* \bigcup F\}.$$

Call Y *positive* if $Y \notin \mathcal{I}(\mathcal{A})$.

Lemma 5.1. *If $\mathcal{A}$ is countably infinite and AD, and $\mathcal{Y}$ is a countable family of $\mathcal{I}(\mathcal{A})$ -positive sets, there is an infinite set a almost disjoint from every member of $\mathcal{A}$ and meeting every $Y \in \mathcal{Y}$ infinitely.*

Proof. Add ω to the requirements. It is positive: a finite subfamily cannot almost cover an infinite AD family. Enumerate $\mathcal{A} = (A_i : i < \omega)$ and the requirements as $(Y_j : j < \omega)$, repeating entries when necessary. At stage n , for every $j \leq n$, choose a fresh point $x_{j,n} \in Y_j \setminus \bigcup_{i \leq n} A_i$, avoiding all previously chosen points. Positivity ensures infinitely many choices at each step. The set of all these points meets each requirement infinitely and each A_i only in the finitely many points selected before stage i . $\square$

Lemma 5.2. *Let $\mathcal{A}$ be AD and C a fin-sequence. If an infinite I satisfies $\bigcup_{n \in I} C_n \in \mathcal{I}(\mathcal{A})$, some infinite $J \subseteq I$ witnesses FI for $\mathcal{A}$ and C .*

Proof. Choose finite $F \subseteq \mathcal{A}$ almost covering that union. Remove the finitely many blocks meeting the finite uncovered part. Every remaining block is contained in $\bigcup F$. There are only finitely many intersection patterns with F , so one nonempty pattern $K \subseteq F$ occurs on an infinite set J of blocks. Members of K have trace J after restriction; members of $F \setminus K$ have empty trace. Each $A \in \mathcal{A} \setminus F$ meets $\bigcup F$ finitely, hence meets only finitely many of the disjoint blocks. The only retained trace is therefore J , and centeredness follows. This is the mechanism of [1, Lemma 2.4]. $\square$

Theorem 5.3 (Known CH construction). *Under CH, there is an infinite FI MAD family of size $\aleph_1$.*

Proof. Start with a countably infinite AD family $(a_n : n < \omega)$. Using CH, enumerate all infinite subsets as $(X_\alpha : \omega \leq \alpha < \omega_1)$ and all fin-sequences as $(C^\alpha : \omega \leq \alpha < \omega_1)$. At stage α , the preceding family $\mathcal{A}_\alpha = \{a_\xi : \xi < \alpha\}$ is countably infinite. Choose an infinite I_α on which every trace of an old member against C^α is finite or cofinite. Successive thinning followed by a diagonal pseudointersection gives such an I_α .

For each already registered $\omega \leq \gamma \leq \alpha$ and finite $F \subseteq \alpha$, put

$$S_\gamma(F) = I_\gamma \cap \bigcap_{\xi \in F} \text{tr}_{C^\gamma}(a_\xi), \quad Y_\gamma(F) = \bigcup_{n \in S_\gamma(F)} C_n^\gamma.$$

Include in the requirement list every such $Y_\gamma(F)$ that is currently $\mathcal{I}(\mathcal{A}_\alpha)$ -positive, together with ω and, if positive, X_α . This list is countable. Lemma 5.1 gives a new member a_α meeting all requirements infinitely and almost disjoint from the preceding family. In particular,

$$Y_\gamma(F) \notin \mathcal{I}(\mathcal{A}_\alpha) \implies |S_\gamma(F \cup \{\alpha\})| = \aleph_0. \quad (2)$$

The implication uses the finiteness and disjointness of the blocks.

The resulting $\mathcal{M} = \{a_\xi : \xi < \omega_1\}$ is AD with distinct infinite members, hence of size $\aleph_1$. It is maximal: at the stage for an infinite X_α , either finitely many preceding members almost cover it, so one already meets it infinitely, or the new member is required to do so.

Fix $C = C^\gamma$. If any infinite union of its blocks belongs to $\mathcal{I}(\mathcal{M})$, Lemma 5.2 gives an FI witness. Otherwise every infinite block union is positive for the final ideal and consequently for every earlier ideal. We show that I_γ itself is a witness. Take finitely many members with infinite traces on I_γ . Those indexed below γ have cofinite traces there, so their finite intersection L is infinite; if there are none, take $L = I_\gamma$. Add the remaining indices in increasing order. Whenever the current intersection L is infinite, its block union is positive in the final ideal, hence positive when the next member was added. Condition (2) preserves infinitude of the intersection. After finitely many steps the intersection of the chosen traces is infinite. Thus all retained traces are centered and $\mathcal{M}$ is FI. $\square$

6 Relative independence and the external model

The invariant $\mathfrak{dp}$ is the least size of the union of two orthogonal families of infinite subsets of ω that cannot be weakly separated. Orthogonality requires finite intersections between the two families. An inseparable cut in an AD family is such a pair, so $\mathfrak{dp} \leq \mathfrak{ap}$. Let $\mathfrak{s}_\omega$ be the least size of a family containing a simultaneous splitter for every countable sequence of infinite subsets of ω . In particular, $\mathfrak{s} \leq \mathfrak{s}_\omega$.

Theorem 6.1 (External consistency input, [2, Theorem 4(3)]). *Relative to the consistency of ZFC, it is consistent that $\mathfrak{s}_\omega = \aleph_1$ and $\mathfrak{ap} = \mathfrak{c} = \aleph_2$.*

This is a weakening of the configuration stated in the cited theorem. Its proof refers to Dow's model and establishes the value of $\mathfrak{s}_\omega$. We use that published consistency result and do not reconstruct its forcing. In this model, the inequalities give

$$\mathfrak{s} = \aleph_1, \quad \mathfrak{ap} = \mathfrak{b} = \mathfrak{a} = \mathfrak{c} = \aleph_2.$$

Thus Theorem 1.1 applies internally to every infinite MAD family.

Theorem 6.2. *If ZFC is consistent, then both $\text{ZFC} + \text{E}$ and $\text{ZFC} + \neg\text{E}$ are consistent. Consequently E is independent of ZFC.*

Proof. For the positive side use the standard relative consistency of CH with ZFC and Theorem 5.3. For the negative side use Theorem 6.1 and the ZFC implication in Theorem 1.1. If either E or its negation were a theorem of ZFC, the respective opposite extension would be inconsistent. $\square$

This is a relative-consistency argument in the usual syntactic sense. It does not assert that ZFC proves its own consistency, and does not assume an extra transitive-model hypothesis. It answers the published existence question [1, Question 3.7]; in the earlier arXiv v2 this is Question 17.

7 Consequences and limits

Corollary 7.1. *ZFC proves $\mathfrak{ap} = \mathfrak{c} \Rightarrow (\text{E} \leftrightarrow \mathfrak{s} = \mathfrak{c})$.*

Proof. If $\mathfrak{s} < \mathfrak{c} = \mathfrak{ap}$, Theorem 1.1 gives $\neg\text{E}$. If $\mathfrak{s} = \mathfrak{c} = \mathfrak{ap}$, apply [1, Theorem 3.4]. $\square$

For an AD family $\mathcal{A}$, the space $\Psi(\mathcal{A})$ has underlying disjoint union $\omega \dot{\cup} \mathcal{A}$. Natural numbers are isolated; a basic neighborhood of $A \in \mathcal{A}$ is $\{A\} \cup (A \setminus F)$ for finite F . Write $\exp(X)$ for the hyperspace of nonempty closed subsets with the Vietoris topology, generated by $U^+ = \{F : F \subseteq U\}$ and $U^- = \{F : F \cap U \neq \emptyset\}$ for open U . Pseudocompactness means that every continuous real-valued function is bounded. Let P assert that some infinite MAD family $\mathcal{A}$ has pseudocompact $\exp(\Psi(\mathcal{A}))$.

Corral and Rodrigues prove $\text{E} \Rightarrow \text{P}$ [1, Proposition 2.5]. They also prove $\mathfrak{ap} = \mathfrak{c} \Rightarrow \text{P}$ [3, Theorem 5].

Corollary 7.2. *If ZFC is consistent, then $\text{ZFC} + \text{P} + \neg\text{E}$ is consistent.*

Proof. The model of Theorem 6.1 satisfies $\mathfrak{ap} = \mathfrak{c}$ and $\mathfrak{s} < \mathfrak{ap}$. Apply the two preceding implications. $\square$

This is stronger than producing one pseudocompact MAD family that is not FI: there is no infinite FI MAD family in that model at all. It does not prove independence of P, nor produce a model of $\neg\text{P}$.

The known sufficient condition $\mathfrak{a} < \mathfrak{s}$ and our necessary condition give

$$\mathfrak{a} < \mathfrak{s} \implies \mathbf{E} \implies \mathfrak{ap} \leq \mathfrak{s}.$$

We do not establish sufficiency of $\mathfrak{ap} \leq \mathfrak{s}$, or even of $\mathfrak{ap} = \mathfrak{s}$. The negative model refutes the implication from $\mathfrak{ap} = \mathfrak{c}$ alone to FI MAD existence asked about in [3, Question 7]; it does not settle the other alternative in that question.

8 Formal verification and provenance

The companion development belongs to the *InfinitaryCombinatorics* library [4]. It reuses its definitions of AD, infinite MAD, fin-sequences, retained traces, centeredness, and FI. Its new modules are in `Formalizations/FIMAD/`. The toolchain remains Lean 4.30.0 with the library’s pinned mathlib revision. A pinned YesMetaZFC dependency supplies the genuine first-order derivation and consistency notions [5].

The checked combinatorial chain proves uniform trace coding, finite-error trace obstruction, and both comparisons $\mathfrak{ap} \leq \mathfrak{b} \leq \mathfrak{a}$. The comparison $\mathfrak{ap} \leq \mathfrak{b}$ is proved directly using binary-tree branches: a weak separator for the eventually-zero branches and a family of infinite branches would eventually bound the increasing enumerations of the latter. The paper’s exact cutoff and exclusion of all infinite FI MAD families therefore have only the stated hypothesis $\mathfrak{s} < \mathfrak{ap}$; no additional cardinal comparison is assumed. Arbitrary labels, including empty positive fibers, are also covered. The development defines $\mathfrak{ap}$ by the least nonseparable AD cardinal and proves that its witness class is nonempty; it does not replace $\mathfrak{ap}$ by an unspecified parameter.

The complete construction in Section 5 is also checked. Starting solely from $2^{\aleph_0} = \aleph_1$, it performs the well-founded recursion on the actual first uncountable ordinal, proves maximality against every infinite subset of ω , and verifies FI against every fin-sequence. The output has size $\aleph_1$. The positive extension, finite ideal-cover, and finite-intersection preservation lemmas are proved within the development, rather than supplied as assumptions.

The development also encodes $\mathbf{E}$ as a closed first-order set-theoretic sentence. Its interpretation uses the model’s own natural numbers, finite sets, functions, and families. A structural proof connects this interpretation to the public YesMetaZFC proof kernel. On ordinary sets, the finite-member formulation of FI is proved equivalent to centeredness of the retained set of traces, including when different members yield the same trace. The cardinals $\mathfrak{dp}$ and $\mathfrak{s}_\omega$ are defined by their actual least witness sizes, their comparisons with $\mathfrak{ap}$ and $\mathfrak{s}$ are proved, and the combinatorial consequences of the BMZ configuration are checked.

Substantive obligations remain. The host proofs on natural numbers have not yet been transported into the set-theoretic models. The BMZ forcing, the positive consistency argument, the stronger positive theorem under $\mathfrak{ap} = \mathfrak{s} = \mathfrak{c}$, the topological results, and the Cohen preservation argument are not formalized here. The concrete ZFC/ $\mathbf{E}$ interface proves independence once actual positive and negative models are supplied; it does not construct those models. Accordingly, Theorem 6.2 remains a paper theorem using the named literature inputs. The source archive includes a statement-by-statement coverage record and the verification procedure.

The research manuscript and this formalization were prepared with generative AI assistance, including GPT-6 Astra. The author supplied the research problem and coordinated the work. Checks performed by the producing AI system do not constitute independent expert review; no

such review of this manuscript is claimed. The author is unaffiliated and received no specific funding for this work. Known positive constructions, cardinal comparisons, forcing results, and hyperspace theorems are credited to their cited sources. No claim of an exhaustive priority search is made.

A The same splitting number and continuum need not decide existence

Corral and Rodrigues construct under CH a FI MAD family that survives adding arbitrarily many Cohen reals; the countable-forcing case is [1, Theorem 5.3] and the arbitrary-Cohen extension is discussed in Section 5 of that paper. This yields a useful comparison, separate from the independence proof above.

Start from GCH with such a family $\mathcal{A}$ of size $\aleph_1$ and force with $\text{Add}(\omega, \omega_2)$. The forcing is ccc and preserves cardinals. Counting nice names gives $\mathfrak{c} \leq \aleph_2$, and the distinct coordinate reals give equality. A nice name for a real or a fin-sequence has countable support, so it appears in a countable-coordinate extension. The cited preservation theorem applies there. An infinite-index FI witness continues to work in the full extension: the family has the same members, and statements about finite intersections of its traces are absolute. The same support argument checks maximality.

The first ω_1 Cohen reals form a splitting family. For any infinite X in the final extension, choose a countable support for its name and then an index below ω_1 outside that support. That coordinate is Cohen over the extension containing X , and the dense requirements to put both bits at arbitrarily large positions in X ensure splitting. Thus $\mathfrak{s} = \aleph_1$. Since the preserved MAD family still has size $\aleph_1$, $\mathfrak{a} = \mathfrak{ap} = \mathfrak{b} = \aleph_1$ as well.

Model	$\mathfrak{s}$	$\mathfrak{ap}$	$\mathfrak{c}$	E
Cohen extension above	$\aleph_1$	$\aleph_1$	$\aleph_2$	true
BMZ model	$\aleph_1$	$\aleph_2$	$\aleph_2$	false

In particular, the hypothesis $\mathfrak{ap} = \mathfrak{c}$ cannot simply be dropped from Corollary 7.1. This appendix relies on the cited preservation result and is not included in the Lean verification.

References

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- [5] *YesMetaZFC*, Lean source repository, <https://github.com/lanxinge/YesMetaZFC>, commit `bae4fcc31b07b505986b11c6c2f13965ae6cd46d`.